AN INFINITE FLAT DESCENT FAMILY WITH THE EQUALIZER PROPERTY BUT NO FINITE SUBFAMILY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Claim I of the canonical-topology section; the companion lemma lemma-Z-not-quasi-compact consumes this family together with Claim II.

Editorial note. The label lemma-non-fpqc-descent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- A nonprincipal ultrafilter U on the set P of prime numbers: every subset of P or its complement lies in U, U is closed under finite intersection, and no finite set lies in U
- For A in U the ring Z-A is the localization of the integers at the multiplicative set generated by the primes in A; its spectrum is the generic point plus the closed primes outside A
- Localization is exact and commutes with tensor products; a torsion abelian group decomposes as the direct sum of its p-primary components
- The equalizer condition for a family of flat ring maps A to A-i on modules: M equals the equalizer of the two natural maps from the product of M tensor A-i to the product over pairs of M tensor A-i tensor A A-j

Source reference. Original lemma 00E8: sources/stacks-chow/tags.txt, source lines 525–525. Prerequisite pointer invoked at 7142; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 090H: sources/stacks-chow/tags.txt, source lines 11410–11410. Prerequisite pointer invoked at 7146; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

the spectrum of the integers as a topological space, because any spectrum is quasi-compact as a topological space (Algebra, Lemma 00E8). No, it refers to the spectrum of the integers in the canonical topology on the category of schemes, and the definition of a quasi-compact object in a site (Sites, Definition 090H).

Let U be a nonprincipal ultrafilter on the set P of prime numbers. For a subset $T \subset P$ we denote $T^c = P \setminus T$ the complement. For $A \in U$ let $S_A \subset \mathbf{Z}$ be the

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multiplicative subset generated by $p \in A$. Set

$$\mathbf{Z}_A = S_A^{-1}\mathbf{Z}$$

Observe that $\mathrm{Spec}(\mathbf{Z}_A) = \{(0)\} \cup A^c \subset \mathrm{Spec}(\mathbf{Z})$ if we think of P as the set of closed points of $\mathrm{Spec}(\mathbf{Z})$. If $A, B \in U$, then $A \cap B \in U$ and $A \cup B \in U$ and we have

$$\mathbf{Z}_{A\cap B} = \mathbf{Z}_A \times_{\mathbf{Z}_{A\cup B}} \mathbf{Z}_B$$

(fibre product of rings). In particular, for any integer n and elements $A_1, \ldots, A_n \in U$ the morphisms

$$\mathrm{Spec}(\mathbf{Z}_{A_1}) \amalg \ldots \amalg \mathrm{Spec}(\mathbf{Z}_{A_n}) \longrightarrow \mathrm{Spec}(\mathbf{Z})$$

factors through $\mathrm{Spec}(\mathbf{Z}[1/p])$ for some p (namely for any $p \in A_1 \cap \ldots \cap A_n$). We conclude that the family of flat morphisms $\{\mathrm{Spec}(\mathbf{Z}_A) \to \mathrm{Spec}(\mathbf{Z})\}_{A\in U}$ is jointly surjective, but no finite subset is.

For a $\mathbf{Z}$-module M we set

$$M_A = S_A^{-1}M = M \otimes_{\mathbf{Z}} \mathbf{Z}_A$$

Claim I: for every $\mathbf{Z}$-module M we have

$$M = \mathrm{Equalizer}\Big(\prod_{A\in U} M_A \rightrightarrows \prod_{A,B\in U} M_{A\cup B} \Big)$$

First, assume M is torsion free. Then $M_A \subset M_P$ for all $A \in U$. Hence we see that we have to prove

$$M = \bigcap\nolimits_{A\in U} M_A \text{ inside } M_P = M \otimes \mathbf{Q}$$

Namely, since U is nonprincipal, for any prime p we have $\{p\}^c \in U$. Also, $M_{\{p\}^c} = M_{(p)}$ is equal to the localization at the prime (p). Thus the above is clear because already $M_{(2)} \cap M_{(3)} = M$. Next, assume M is torsion. Then we have

$$M = \bigoplus\nolimits_{p\in P} M[p^\infty]$$

and correspondingly we have

$$M_A = \bigoplus\nolimits_{p\notin A} M[p^\infty]$$

because we are localizing at the primes in A. Suppose that $(x_A) \in \prod M_A$ is in the equalizer. Denote $x_p = x_{\{p\}^c} \in M[p^\infty]$. Then the equalizer property says

$$x_A = (x_p)_{p\notin A}$$

and in particular it says that x_p is zero for all but a finite number of $p \notin A$. To finish the proof in the torsion case it suffices to show that x_p is zero for all but a finite number of primes p. If not write $\{p \in P \mid x_p \neq 0\} = T \amalg T'$ as the disjoint union of two infinite sets. Then either $T \notin U$ or $T' \notin U$ because U is an ultrafilter (namely if both T, T' are in U then U contains $T \cap T' = \emptyset$ which is not allowed). Say $T \notin U$. Then $T = A^c$ and this contradicts the finiteness mentioned above. Finally, suppose that M is a general module. Then we look at the short exact sequence

$$0 \to M_{tors} \to M \to M/M_{tors} \to 0$$

and we look at the following large diagram

$$\begin{array}{ccccc}
M_{tors} & \longrightarrow & \prod_{A\in U} M_{tors,A} & \rightrightarrows & \prod_{A,B\in U} M_{tors,A\cup B} \\
\downarrow & & \downarrow & & \downarrow \\
M & \longrightarrow & \prod_{A\in U} M_A & \rightrightarrows & \prod_{A,B\in U} M_{A\cup B} \\
\downarrow & & \downarrow & & \downarrow \\
M/M_{tors} & \longrightarrow & \prod_{A\in U}(M/M_{tors})_A & \rightrightarrows & \prod_{A,B\in U}(M/M_{tors})_{A\cup B}
\end{array}$$

Doing a diagram chase using exactness of the columns and the result for the torsion module M_{tors} and the torsion free module M/M_{tors} proving Claim I for M.

Lemma 0.1. *There exists a ring A and an infinite family of flat ring maps $\{A \to A_i\}_{i\in I}$ such that for every A-module M*

$$M = \mathit{Equalizer}\Big(\prod_{i\in I} M \otimes_A A_i \rightrightarrows \prod_{i,j\in I} M \otimes_A A_i \otimes_A A_j \Big)$$

but there is no finite subfamily where the same thing is true.

Proof. See discussion above. □